

Moment Inequalities and Partial Identification¹

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Grad IO

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Summary

Overview

In this lecture we explore recent approaches to estimating models through the use of inequality restrictions, rather than equality restrictions.

The motivation for these approaches comes from:

- ▶ Economic problems that cannot be empirically investigated with more standard techniques (e.g., discrete games)
- ▶ Sampling methodology or measurement limitations in data (e.g. we don't know an agent's exact income, but we know whether it lies in an interval, so we can get inequalities on income; or we have topcoded income)

- ▶ The assumptions required in the latter case to ensure that moment inequalities estimators have desirable properties are quite straightforward.
- ▶ The assumptions required in the former case are more subtle.
- ▶ Once we investigate these assumptions, we may start questioning the assumptions underlying the estimators of more traditional behavioral problems (particularly those that require discrete actions in a game or assume full information about all players, etc.).

- ▶ The appeal of using moment inequalities to analyze a behavioral model is that the set up for the econometric problem is often the same as the set up for a theoretical model.
- ▶ This similarity makes it easy to interpret the results in a way that is consistent with the theory.
- ▶ However, dealing with unobservables can be tricky in this context.

Economic Model

- ▶ We know how to estimate models based on moment conditions like:

$$E[g(z_i, \theta)] = 0$$

- ▶ But instead we might form moments like:

$$E[h(z_i, \theta)] \leq 0$$

either in addition to our moments that hold with equality, or as a replacement for.

- ▶ Instead of identifying a single point $\hat{\theta}$ our restrictions might only **identify a set** $S(\Theta)$.
- ▶ Both estimation and inference are going to work differently in this world.

- ▶ The econometrician observes a set of choices made by various agents.
- ▶ Assume agents expect the choices they made to lead to returns that are higher than the returns they would have earned had they made an alternative feasible choice.
- ▶ Assume a parametric *return function*.
- ▶ For each value of θ , compute the difference between the observable part of the actual realized returns and the observable part of returns the agents would have earned had they made the alternative choice.

- ▶ Estimator: Accept any value of θ that, on average, makes the observed decisions better (more profitable) than the alternative.
- ▶ Question: When do such (possibly set valued) estimators enable us to make valid inferences on the parameters of interest?

There is more than one set of conditions that can be used to justify inequality estimators. We will consider the following two such sets of conditions:

- ▶ Those that use only “structural errors” (i.e., assumptions that are the multiple-agent analog of the assumptions used in discrete choice econometrics).
- ▶ Those that incorporate expectational and measurement errors along with the structural errors.

- ▶ Ciliberto and Tamer (*Econometrica* 2009) develop the estimator based on the first set of conditions in the context of an entry model.
- ▶ Pakes, Porter, Ho and Ishii (*Econometrica* 2015) develop the estimator based on the second set of conditions.
- ▶ Pakes (*Econometrica* 2010) provides a comparison of the two sets of conditions.

Conditions Common to the Two Inequality Estimators

Four Conditions Required for Discrete Games

- ▶ Let $\pi(\cdot)$ be the payoffs,
- ▶ d_i and $\mathbf{d}_{-i}$ be the agent's and its competitors' choices
- ▶ $\mathbf{y}_i$ be any variable (other than the decision variables) that affects the agent's payoffs
- ▶ D_i be the choice set, and Ω_i be the agent's information set.
- ▶ $\varepsilon[\cdot | \Omega_i]$ is the agent's expectation conditional on the information set Ω_i .

Nash Condition (C1)

$$\sup_{d \in D_i, d \neq d_i} \varepsilon [\pi(d, \mathbf{d}_{-i}, \mathbf{y}_i, \theta_0) | \Omega_i] \leq \varepsilon [\pi(d_i, \mathbf{d}_{-i}, \mathbf{y}_i, \theta_0) | \Omega_i]$$

where $D_i \subset D$, for $i = 1, \dots, n$.

- ▶ No restriction on choice set; could be discrete (e.g., all bilateral contracts) or continuous (if so, the optimum could be at a corner and the objective function may have non-convexities), or a combination of discrete/continuous (e.g. a carbon tax on gas consumption may affect consumers' choices of car and the number of miles traveled conditional on car choice).
- ▶ C1 is a necessary condition for a Nash equilibrium, and is meant to be a rationality assumption (i.e., the agent's choice is optimal wrt his beliefs). However, it does not imply uniqueness, and equilibrium selection can differ across observations.
- ▶ To check the Nash Condition, we need an approximation to what profits *would have been* had the agent made a choice which in fact he did not make. This requires a model of how the agent thinks that $\mathbf{d}_{-i}$ and $\mathbf{y}_i$ are likely to change in response to a change in the agent's decision.

Counterfactual Condition (C2)

The model for how the agent thinks $(\mathbf{y}_i, \mathbf{d}_{-i})$ are likely to respond to changes in d_i may depend on other variables, say z_i , but we require z_i to be exogenous (i.e., the agent believes z_i will not change if the agent changes its own decision).

Condition C2 formalizes this assumption.

$$\mathbf{d}_{-i} = d^{-i}(\mathbf{d}_i, \mathbf{z}_i), \mathbf{y}_i = y(\mathbf{z}_i, \mathbf{d}_i, \mathbf{d}_{-i}), \text{ and}$$

the distribution of $\mathbf{z}_i$ conditional on I_i does not depend on d_i .

- ▶ If we have simultaneous moves then $d^{-i}(d', \mathbf{z}_i) = \mathbf{d}_{-i}$ (i.e., there is no need for an explicit model of reactions by competitors, and Condition C2 is satisfied.)
- ▶ If there are sequential decisions and we want to use the decision of the first player in the analysis, then we have to specify a model for what the first player thinks the second player would do were the first player to change his decision.
- ▶ If there is a $\mathbf{y}$ which is “endogenous” - i.e. its distribution depends on d_i - then we need a model of that dependence.

Implications of conditions C1 and C2:

If $d' \in D_i$ and

$$\Delta\pi(d_i, d', d_{-i}, z_i) = \pi(d_i, d_{-i}, z_i) - \pi(d', d_{-i}, z_i)$$

then

$$\mathbb{E}[\Delta\pi(d_i, d', \mathbf{d}_{-i}, \mathbf{z}_i) | \Omega_i] \geq 0.$$

To use this inequality directly as a basis for an estimation algorithm, we need the relationships between:

- ▶ The expectations underlying agents' decisions ($\mathbb{E}(\cdot)$) and the expectations of the observed sample moments ($E(\cdot)$),
- ▶ $\pi(\cdot, \theta)$ and (z_i, d_i, d_{-i}) and their observable analogues.

This is where the two approaches differ. We covered the first set in the last lecture.

Entry Models with Structural Errors, Example

Before specifying the next two conditions, consider the information structure of a simple 2-firm entry model.

Let $a_i \in \{0, 1\}$ denote the action of player $i = 1, 2$. The profits are given by:

$$\Pi_i(s) = \begin{cases} \beta' s - \delta a_{-i} + \epsilon_i, & \text{if } a_i = 1 \\ 0 & \text{otherwise} \end{cases}$$

where s denotes market-level control variables.

Firm entry choices are interdependent, in the sense that firm 1's profits from entering (and, hence, his decision to enter) depend on whether firm 2 is in the market.

As before, the error terms ϵ_i are assumed to be observed by both firms, but not by the econometrician. This is a “perfect information” game. (Seim (2006) considers an “incomplete information” game.)

Entry Models with Structural Errors, Example

For fixed values of the errors $\epsilon \equiv (\epsilon_1, \epsilon_2)$ and parameters $\theta \equiv (\alpha_1, \alpha_2, \beta_1, \beta_2)$, the Nash equilibrium values a_1^*, a_2^* must satisfy best-response conditions. For fixed (θ, ϵ) , the best-response conditions are:

$$a_1^* = 1 \Leftrightarrow \Pi_1(a_2^*) \geq 0$$

$$a_1^* = 0 \Leftrightarrow \Pi_1(a_2^*) < 0$$

$$a_2^* = 1 \Leftrightarrow \Pi_2(a_1^*) \geq 0$$

$$a_2^* = 0 \Leftrightarrow \Pi_2(a_1^*) < 0$$

For some values of parameters, there may be multiple equilibria. Define the mutually exclusive outcome indicators:

$$Y_1 = 1(a_1 = 1, a_2 = 0)$$

$$Y_2 = 1(a_1 = 0, a_2 = 1)$$

$$Y_3 = 1(a_1 = 0, a_2 = 0)$$

$$Y_4 = 1(a_1 = 1, a_2 = 1)$$

Entry Models with Structural Errors, Example

The moment inequalities for the Nash equilibrium assumptions are:

- $[1 - \Phi(-\beta' s)][\Phi(\delta - \beta' s)] \geq E[Y_1 | s] \geq [1 - \Phi(-\beta' s)]\Phi(-\beta' s) + [1 - \Phi(\delta - \beta' s)][\Phi(\delta - \beta' s) - \Phi(-\beta' s)]$
- $[1 - \Phi(-\beta' s)][\Phi(\delta - \beta' s)] \geq E[Y_2 | s] \geq [1 - \Phi(-\beta' s)]\Phi(-\beta' s) + [1 - \Phi(\delta - \beta' s)][\Phi(\delta - \beta' s) - \Phi(-\beta' s)]$
- $[\Phi(-\beta' s)]^2 \geq E[Y_3 | s] \geq [\Phi(-\beta' s)]^2$
- $[1 - \Phi(\delta - \beta' s)]^2 \geq E[Y_4 | s] \geq [1 - \Phi(\delta - \beta' s)]^2$

Entry Models with Structural Errors, Example

We've already seen one alternative remedy to this problem.

- ▶ Instead of modeling events $Y_1 = 1$ and $Y_2 = 1$ separately ...
- ▶ Model the aggregate event $Y_5 \equiv Y_1 + Y_2 = 1$, which is the event that *only one firm* enters.
- ▶ In other words, just model the likelihood of *number of entrants*, but not the identities of entrants. This was done in Berry's (1992) paper.
- ▶ More recently, researchers attempt to more flexibly address the problem of multiple equilibria by:
 1. Generalizing the information structure of the game
 2. Working directly with the moment inequalities. How these models are taken to data differ based on the assumptions that the researcher puts on the model's sources of error (and on the information structure of the game).

Approach #1: No Specification Error

Early papers focus on games where the moment (in)equalities are generated by “structural” errors only (i.e. those observed by firms, but not by the econometrician).

- ▶ Early versions of these models select an equilibrium ex-ante: Bresnahan and Reiss (1991 and others), Berry (1992), Mazzeo (2003), and Seim (2006).
- ▶ Ciliberto and Tamer (2009) follow this approach too, but allow for multiple equilibria. As a result of this, the parameters of their model are “set identified,” as we will see later on.

Each of these models are Full Information models except Seim; she introduces Asymmetric Information between firms but otherwise assumes no specification error.

Approach #2: Expectational and Measurement Error

- ▶ More recently, estimation of discrete games has evolved to consider the case where the moment (in)equalities are generated by non-structural, expectational errors, which are not known by agents at the time that their decisions are made.
- ▶ This approach is broader than the entry literature per se, and follows the approach taken in Pakes, Porter, Ho, and Ishii.

**Extra Conditions under Full
Information and No Specification
Error**

Expectational Condition (FC3)

$$\pi(d, d_{-i}, z_i, \theta_0) = \varepsilon [\pi(d, \mathbf{d}_{-i}, \mathbf{z}_i, \theta_0) | \Omega_i] \quad \forall \quad d \in D_i$$

- ▶ FC3 implies that the model does not allow for any expectational error. That is, it rules out asymmetric and/or incomplete information.
- ▶ The first assumption ensures that other agents' actions (d_{-i}) are known with certainty at the time the agent makes its decision, and the second ensures that the agent knows z_i with certainty at the time its decision is made.
- ▶ Note this restricts D_i to pure strategies. At a cost of notational complexity the model could be augmented to account for sequential games.

$\pi(., \theta)$ is known.

$z_i = (v_{2,i}^f, z_i^o)$, $(d_i, d_{-i}, z_i^o, z_{-i}^o)$ observed,

$(v_{2,i}^f, v_{2,-i}^f) | z_i^o, z_{-i}^o \sim F(., \theta)$, $F(., \theta)$ is known.

- ▶ FC4 says that the model does not allow for any specification or measurement error.
- ▶ Our functional form for the profit equation is exactly the same as that of the agents.
- ▶ Some of the z_i are observed by the econometrician (the z_i^o) and some are not ($v_{2,i}^f$).
- ▶ The agents know $(v_{2,i}^f, v_{2,-i}^f)$ (from FC3), though the econometrician does not.
- ▶ The econometrician knows their joint distribution.
- ▶ We will see below that these assumptions provide a lot of power.

Implications of FC3 and FC4

Substituting FC3 and FC4 into the moment inequality gives:

$$\Delta\pi(d_i, d', d_{-i}, z_i^o, v_{2,i}^f; \theta_0) \geq 0,$$

$\forall d \in D_i$, and

$$(v_{2,i}^f, v_{2,-i}^f) | z_i^o, z_{-i}^o \sim F(\cdot; \theta_0).$$

This is almost enough to build an estimation algorithm. It does leave the logical problem that there may not be a θ that satisfies these conditions for all vectors of decisions. To ensure that the model assigns positive probability to the observed decisions for some θ we typically also assume additive separability:

$$\pi(d_i, d_{-i}, z_i^o, v_{2,i}^f) = \pi^{as}(d, d_{-i}, z_i^o, \theta_0) + v_{2,i,d}^f,$$

and that the distribution $v_{2,i}^f$ conditional on $v_{2,-i}^f$ has full support.

Notes:

- ▶ The additive separability of $v_{2,i,d}^f$ cannot be obtained definitionally, by assuming v_2 is a residual from a projection because the RHS contains a decision variable which depends on v_2 . In the single-agent discrete choice literature we can solve out for the d_i to obtain a function that depends only on “exogenous” variables. Here we can’t because d_{-i} is on the right hand side, and by assumption the $-i$ agents know $v_{2,i}$ when making their decisions.
- ▶ Early work on entry looked for a useful reduced form (one that could be used to summarize the effects of environmental characteristics of the market on number of participating agents). It typically assumed orthogonality of the error and solved for the optimal decision of each agent (enter or not). This work tended to find that the implied profits increased with the number of competitors. This was because more firms entered in more profitable markets (alternatively the error had components that were common to all participating agents, and hence were correlated with d_{-i}).

- ▶ Although the usual reduced-form assumptions used to generate discrete choice models do not do well when there are interacting agents, there is always a reduced-form for the single agent model that does make sense. (Regress profits on variables of interest, assume a conditional distribution of the error, compute the choice as a function of the error, and form a standard estimator.) It is the fact that this does not work for multiple agent problems that lead to the developments below.
- ▶ Suppose we wanted a reduced form for our problem. We could regress $\pi^{as}(\cdot)$ on variables of interest (e.g. d_{-i} and other things). Were we to do so, we would pick up an additional error which is by construction, orthogonal to the included variables. Then we would have to deal with both errors in estimation, and they have different properties. The models we describe next, are going after such a reduced form, but they do not allow for the latter error. So there is a question of how any logical inconsistency affects the results.

Estimation from Inequality Conditions (Structural Errors Only)

This section follows Ciliberto and Tamer (2009), which studies entry in the airline industry.

Start with the formulation of profit in Berry (1992):

$$\Pi_{m,k,N} = X_m\beta - \delta\log N + Z_k\alpha + (\rho u_{m0} + \sqrt{1 - \rho^2}u_{mk})$$

- ▶ $(X_m\beta - \delta\log N)$ is the observable component of profit that is common to all firms in a market
- ▶ $Z_k\alpha$ is the observable component of profit that varies across firms within a market.
- ▶ The unobserved components of profit also have a common component (ρu_{m0}) and a firm-specific component $(\sqrt{1 - \rho^2}u_{mk})$.

Estimation from Inequality Conditions (Structural Errors Only)

Ciliberto and Tamer allow for a more flexible profit function, seen here:

$$\Pi_{m,i} = S'_m \alpha + Z'_{im} \beta_i + W'_{im} \gamma_i + \sum_{j \neq i} \delta_j^i y_{jm} + \sum_{j \neq i} Z'_{jm} \phi_j^i y_{jm} + \epsilon_{im}$$

- ▶ S_m denotes market characteristics common to all firms
- ▶ Z_{im} is a vector of firm characteristics that enter into the profits of all firms in the market (e.g., product attributes that consumers value)
- ▶ W_{im} is a vector of firm characteristics that only affect firm i 's profit in market m
- ▶ y_{jm} is an indicator for the presence of other firms $j \neq i$ in market m
- ▶ δ_j^i captures the effect of having firm j in market m on firm i 's profit
- ▶ ϕ_j^i captures firm-interaction effects that arise through the Z_m 's.
- ▶ ϵ_{im} has several components: destination, origin, airport, and a firm-market unobservable. This maps to the $v_{2,i}$ error in our notes, and is assumed to be observed by all players.

Estimation from Inequality Conditions (Structural Errors Only)

We can write down the conditions from the theoretical model (C1 and C2) given this profit function, as well as the conditions that allow us to translate the theoretical model to the data in a perfect information, “structural errors” world (FC3 and FC4).

Note that although the model does specify a parametric distribution for the $(v_{2,i}^f, v_{2,-i}^f)$ conditional on the observables, it does not deliver a likelihood.

There is no function that takes values of the z^o , v_2 and θ vectors into uniquely-specified actions, so we cannot construct probabilities for those actions.

This result comes directly from the inequalities of the entry game, and is due to the possibility of multiple equilibria.

1. We can check whether the conditions of the model are satisfied at the observed (d_i, d_{-i}) for any $(v_{2,i}^f, v_{2,-i}^f)$ and θ , and this, together with $F(., \theta)$, enables us to calculate the probability of those conditions being satisfied at any θ .

These are necessary conditions for the choice to be made: therefore when $\theta = \theta_0$ the probability of satisfying them must be weakly greater than the probability of observing (d_i, d_{-i}) .

I.e. the model delivers an *outer measure* for the actions conditional on θ .

2. We can check whether (d_i, d_{-i}) are the only values of the decision variables to satisfy the necessary conditions for any $(v_{2,i}^f, v_{2,-i}^f)$ and θ , and this can be used to provide a lower bound on the probability of actually observing (d_i, d_{-i}) given θ .

When $\theta = \theta_0$ the probability that we observe these decisions is weakly greater than the probability that they're the only values that are consistent with the model.

I.e., the model also delivers an *inner measure* for the action conditional on θ .

More formally, define the probability that the model generated by C1, C2, FC3 and FC4 with the additive separability constraint, a model we will call MF, is satisfied at a particular (d_i, d_{-i}) for a given θ to be:

$$\bar{P} \{ (d_i, d_{-i}) | \theta \} = \mathbb{P} \left\{ (v_{2,i}^f, v_{2,-i}^f) : (d_i, d_{-i}) \text{ satisfy MF} | z_i^o, z_{-i}^o, \theta \right\},$$

and the analogous lower bound to be:

$$\underline{P} \{ (d_i, d_{-i}) | \theta \} = \mathbb{P} \left\{ (v_{2,i}^f, v_{2,-i}^f) : \text{only } (d_i, d_{-i}) \text{ satisfy MF} | z_i^o, z_{-i}^o, \theta \right\}.$$

Were we to know the equilibrium selection mechanism we could also calculate the actual likelihood of (d_i, d_{-i}) for a given θ or

$$P \{(d_i, d_{-i}) | \theta\} = \mathbb{P} \{(d_i, d_{-i}) | z_i^o, z_{-i}^o, \theta\}.$$

We *do not know* the selection mechanism but we do know that for the true selection mechanism when $\theta = \theta_0$

$$\boxed{\bar{P} \{(d_i, d_{-i}) | \theta\} \geq P \{(d_i, d_{-i}) | \theta\} \geq \underline{P} \{(d_i, d_{-i}) | \theta\} .}$$

Let $\{\}$ be the indicator function which takes the value one if the condition inside the brackets is satisfied and zero elsewhere.

Let $h(\cdot)$ be a function which takes only positive values.

Let $E(\cdot)$ provide expectations conditional on the process actually generating the data (including the equilibrium selection process).

Then MF implies that

$$\begin{aligned} & E_{v_2}(N^{-1} \sum_i (\bar{P}\{(d_i, d_{-i})|\theta\} - \{d = d_i, d^{-i} = d_{-i}\})h(z_i^o, z_{-i}^o)) \\ &= \boxed{N^{-1} \sum_i (\bar{P}\{(d_i, d_{-i})|\theta\} - P\{(d_i, d_{-i})|\theta\})h(z_i^o, z_{-i}^o) \geq 0 \text{ at } \theta = \theta_0.} \end{aligned}$$

which gives us a moment inequality. An analogous moment inequality can be constructed from the condition on $\underline{P}\{(d_i, d_{-i})|\theta\}$:

$$= \boxed{N^{-1} \sum_i (P\{(d_i, d_{-i})|\theta\} - \underline{P}\{(d_i, d_{-i})|\theta\})h(z_i^o, z_{-i}^o) \geq 0 \text{ at } \theta = \theta_0.}$$

Ciliberto and Tamer 2009

The estimation routine:

- ▶ constructs unbiased estimates of $(P(\cdot|\theta), \bar{P}(\cdot|\theta))$,
- ▶ substitutes them for the true values of the probability bounds into these moments, and
- ▶ accepts values of θ for which the moment inequalities are satisfied.

Since typically neither the upper nor the lower bound are analytic functions of θ , we employ simulation techniques to obtain an unbiased estimate of them.

This process can be computationally expensive, often too computationally burdensome to do.

We can get away with fewer function evaluations if we want to rely only on the upper bound probability $\bar{P}(\cdot|\theta)$, as then we can drop as many comparisons as we want, though by dropping inequalities you are likely to widen the set of θ the model accepts.

The same data used in Berry (1992):

- ▶ DB1B data from 2001 (no price data).
- ▶ A market is a trip between two airports and a direction, regardless of stops.
- ▶ The data covers a sample of markets between the top 100 MSA's; 2742 markets.
- ▶ Ciliberto and Tamer focus on strategic interactions between American, Delta, United, and Southwest and pay particular attention to Dallas because of the Wright Amendment.
- ▶ They construct Berry's "airport presence" variable; they do not have cost data, but they have plane capacities and they use that to construct an opportunity fixed cost of serving a market.

Results - Ciliberto and Tamer (2009)

Tables 3 and 4 in Ciliberto and Tamer report confidence intervals for points (i.e. the probability that the confidence set covers the identified set is .95). Table 3 assumes iid errors.

- ▶ The first column constrains competitive effects of each firm's presence in the market on other firms to be the same. So, as in Berry (1992) there is a unique number of firms. The competitive effects (number of other airlines serving the city-pair) and the airport presence effect are both strong, as in prior work.
- ▶ Column 2 allows competitive effects to vary across firms. No longer a unique number of firms (depends on who enters). Competitive effects are similar except for the low-cost carriers which have a bigger impact, and airport presence stronger yet.
- ▶ Column 4 allows one airline's presence to affect different airlines differently. Now Large airlines (LAR) and Southwest (=WN) have strongly negative effects on LCC, and there are smaller differences among the rest.

Results - Ciliberto and Tamer (2009)

TABLE III
EMPIRICAL RESULTS^a

	Berry (1992)	Heterogeneous Interaction	Heterogeneous Control	Firm-to-Firm Interaction
Competitive fixed effect	[-14.151, -10.581]			
AA		[-10.914, -8.822]	[-9.510, -8.460]	
DL		[-10.037, -8.631]	[-9.138, -8.279]	
UA		[-10.101, -4.938]	[-9.951, -5.285]	
MA		[-11.489, -9.414]	[-9.539, -8.713]	
LCC		[-19.623, -14.578]	[-19.385, -13.833]	
WN		[-12.912, -10.969]	[-10.751, -9.29]	
LAR on LAR				
LAR: AA, DL, UA, MA				[-9.086, -8.460]
LAR on LCC				[-20.929, -14.578]
LAR on WN				[-10.294, -9.29]
LCC on LAR				[-22.842, -9.29]
WN on LAR				[-9.093, -7.13]
LCC on WN				[-13.738, -7.13]
WN on LCC				[-15.950, -11.457]
Airport presence	[3.052, 5.087]	[11.262, 14.296]	[10.925, 12.541]	[9.215, 10.414]
Cost	[-0.714, 0.024]	[-1.197, -0.333]	[-1.036, -0.373]	[-1.060, -0.373]
Wright	[-20.526, -8.612]	[-14.738, -12.556]	[-12.211, -10.503]	[-12.092, -10.503]
Dallas	[-6.890, -1.087]	[-1.186, 0.421]	[-1.014, 0.324]	[-0.975, 0.291]
Market size	[0.072, 2.247]	[0.522, 1.245]	[0.372, 0.960]	[0.044, 0.960]

Results - Ciliberto and Tamer (2009)

TABLE III—Continued

	Berry (1992)	Heterogeneous Interaction	Heterogeneous Control	Firm-to-Firm Interaction
Market distance	[4.356, 7.046]	[0.106, 1.002]	[0.062, 0.627]	[−0.057, 0.000]
WN			[−2.441, −1.121]	
LCC			[−0.714, 1.858]	
Close airport	[4.022, 9.831]	[−0.769, 2.070]	[−0.289, 1.363]	[−1.399, −0.000]
WN			[1.751, 3.897]	
LCC			[0.392, 5.351]	
U.S. center distance	[1.452, 3.330]	[−0.932, −0.062]	[−0.275, 0.356]	[−0.606, 0.000]
WN			[−0.357, 0.860]	
LCC			[−1.022, 0.673]	
Per capita income	[0.568, 2.623]	[−0.080, 1.010]	[0.286, 0.829]	[0.272, 1.000]
Income growth rate	[0.370, 1.003]	[0.078, 0.360]	[0.086, 0.331]	[0.094, 0.000]
Constant	[−13.840, −7.796]	[−1.362, 2.431]	[−1.067, −0.191]	[0.381, 2.000]
MA			[−0.016, 0.852]	
LCC			[−2.967, −0.352]	
WN			[−0.448, 1.073]	
Function value	1756.2	1644.1	1627	1658.3
Multiple in identity	0.837	0.951	0.943	0.969
Multiple in number	0	0.523	0.532	0.536
Correctly predicted	0.328	0.326	0.325	0.308

Results - Ciliberto and Tamer (2009)

TABLE IV
VARIABLE COMPETITIVE EFFECTS

	Independent Unobs	Variance-Covariance	Only Costs
Fixed effect			
AA	[−9.433, −8.485]	[−8.817, −8.212]	[−11.351, −9.686]
DL	[−10.216, −9.255]	[−9.056, −8.643]	[−12.472, −11.085]
UA	[−6.349, −3.723]	[−4.580, −3.813]	[−10.671, −8.386]
MA	[−9.998, −8.770]	[−7.476, −6.922]	[−11.906, −10.423]
LCC	[−28.911, −20.255]	[−14.952, −14.232]	[−11.466, −8.917]
WN	[−9.351, −7.876]	[−6.570, −5.970]	[−12.484, −10.614]
Variable effect			
AA	[−5.792, −4.545]	[−4.675, −3.854]	
DL	[−3.812, −2.757]	[−3.628, −3.030]	
UA	[−10.726, −5.645]	[−8.219, −7.932]	
MA	[−6.861, −4.898]	[−7.639, −6.557]	
LCC	[−9.214, 13.344]		
WN	[−10.319, −8.256]	[−11.345, −10.566]	
Airport presence	[14.578, 16.145]	[10.665, 11.260]	

Results - Ciliberto and Tamer (2009)

TABLE IV-Continued

Wright	[−17.800, −16.346]	[−16.781, −15.357]	[−14.284, −10.479]
Dallas	[0.368, 1.323]	[0.839, 1.132]	[−5.517, −2.095]
Market size	[0.230, 0.535]	[0.953, 1.159]	[1.946, 2.435]
WN	[0.260, 0.612]	[0.823, 1.068]	
LCC	[−0.432, 0.507]		
Market distance	[0.009, 0.645]	[0.316, 0.724]	[−0.039, 1.406]
WN	[−3.091, −1.819]	[−2.036, −1.395]	
LCC	[−1.363, 1.926]		
Close airport	[−0.373, 0.422]	[0.400, 1.433]	[3.224, 6.717]
WN	[1.164, 3.387]	[2.078, 2.450]	
LCC	[1.059, 3.108]	[1.875, 2.243]	
U.S. center distance	[−9.271, 0.506]	[0.015, 0.696]	[2.346, 3.339]
WN	[0.276, 1.008]	[0.668, 1.097]	
LCC	[−0.930, 0.367]		
Per capita income	[0.929, 1.287]	[0.824, 1.052]	[1.416, 2.307]
Income growth rate	[0.136, 0.331]	[0.151, 0.316]	[1.435, 2.092]
Constant	[−0.522, 0.163]	[−0.827, −0.523]	[−12.404, −10.116]

Ho 2009

- ▶ Theory testing
- ▶ Measurement
- ▶ Methodology

► Theory testing

- Can a bargaining model explain the hospital-insurance plan contracting process, rationalizing the observed network of hospital-plan relationships?

► Measurement

- What characteristics of hospitals and plans explain the level of surplus hospitals can extract from the relationship?
- What is the effect of capacity constraints on producer welfare? Might the level of capacity be a relevant choice variable for a profit-maximizing firm?

► Methodology

- What assumptions are needed on behavior to develop a moment inequality estimator for static contracting problems?
- What can information on ex-post network formation reveal about private negotiated prices?

Main Idea

- ▶ Model demand for hospitals and health plans, accounting for the hospital network of each plan in the consumer's plan choice
- ▶ Model the supply side negotiation between hospitals and plans in forming equilibrium networks, which determines the division of profits
- ▶ To increase their share of the surplus from contracting, hospitals have incentives to:
 - Invest in quality to attract more patients, lower costs
 - Merge with other providers, to improve bargaining position
 - Under-invest in capacity

Main Idea

► Findings

- “Star” hospitals capture \$6700 more per patient than other providers, on costs of \$11,000
- Hospitals with capacity constraints have markups of \$6900 per patient more than those without constraints
- System hospitals have \$180,000/month greater profits than other providers

Data

- ▶ Insurer plan data cover all managed care insurers in 43 major markets across the US for Q3, Q4 of 2002 (cross-section)
 - Premiums earned, number of enrollees, tax status of each carrier
 - Data on clinical performance and patient satisfaction with health plans
- ▶ Hospital data from Medstat from private insurers; includes encounter-level data on hospital admissions during 2 year period.
 - Patient's diagnosis and characteristics, identity of hospital, type of plan
 - Hospital characteristics from AHA
- ▶ Data on network of hospitals for every HMO/POS plan in every market considered in March/April 2003

Model: Stages

1. Plans choose quality and products; Hospitals choose capacity, location, product mix, system mergers.
2. Hospitals make simultaneous take-it-or-leave-it price offers to all plans in the market
3. Plans choose whether to accept these offers, forming their hospital network
4. Plans set premiums to maximize profits after a change in networks
5. Consumers and employers jointly choose plans
6. Sick consumers visit hospitals; plans pay hospitals per service provided.

Model: Hospital Demand

$$u_{i,h,l} = \eta_h + x_h \alpha + x_h \nu_{i,l} \beta + \varepsilon_{i,h,l}$$

- ▶ individual i , hospital h , diagnosis l
- ▶ x_h observed hospital characteristics
- ▶ $\nu_{i,l}$ observed characteristics of consumers
- ▶ Estimate via ML, using Medstat data
 - Medstat doesn't have hospital networks for managed care enrollees; use only data on indemnity and PPO enrollees whose choice set is unrestricted
 - Assume: indemnity/PPO enrollees have same preferences over hospitals as managed care enrollees (vertical preferences)

Model: Health Plan Demand

$$\tilde{u}_{ijm} = \xi_{jm} + z_{jm}\lambda + \gamma_1 EU_{ijm} + \gamma_2 \frac{prem_{j,m}}{y_i} + \omega_{ijm}$$

- ▶ individual i, plan j, market m
- ▶ (z_{jm}, ξ_{jm}) observed and unobserved plan characteristics
- ▶ outside option = choosing to be uninsured; indemnity/PPO is separate choice in each mkt
- ▶ IV for premium
 - plan char, avg hourly hospital wage, avg weekly nurse wage
 - exclusion restriction: health plan costs correlated with premiums but not with unobs plan quality
- ▶ Find: consumers value EU from network in plan choice

Model: Producer surplus generated by network

$$S_{j,m}(H_j, H_{-j}) = \sum_i (n_i s_{ijm}(H_j, H_{-j}) [prem_{j,m} - p_i \sum_{h \in H_j} s_{i,h}(H_j) cost_h])$$

- ▶ The shares $s_{ijm}(H_j, H_{-j})$ are plan j's predicted shares of type i people when networks (H_j, H_{-j}) offered (flow of consumers to plans after network changes)
- ▶ $s_{i,h}(H_j)$ hospitals h's predicted share of type i people (flow of consumers to hospitals after network change)

Model: Producer surplus generated by network

$$S_{j,m}(H_j, H_{-j}) = \sum_i (n_i s_{i,j,m}(H_j, H_{-j}) [prem_{j,m} - p_i \sum_{h \in H_j} s_{i,h}(H_j) \text{cost}_h])$$

- ▶ Premiums adjust in response to changes in hospital network
 - (1) Estimate supply model assuming fixed premiums
 - (2) Allow all plans to simultaneously adjust their premiums to max profits
 - Comment: with panel data, could push further
- ▶ No non-hospital variable costs
- ▶ Adjusts for capacity constraints at 85% level

Model: Negotiation

- ▶ All hospitals make TIOLI offers of {contract,null offer}
- ▶ All plans simultaneously respond
- ▶ Offers are private info; plans have passive beliefs (if plan gets an alternative offer from h, doesn't change plan's beliefs about offers h makes to its competitors)

$$\pi_{j,m}^P = S_{j,m}(H_j, H_{-j}) - c_{j,m}^{Hosp}(H_j, H_{-j}, X, \theta) - c_{j,m}^{nonhosp}(H_j, H_{-j}, X, \theta)$$

$$\begin{aligned}\pi_{j,m}^{P,o}(\cdot) &= \pi_{j,m}^P + \mu_{j,H_j} \\ E[\pi_{j,m}^P(H_j, H_{-j}, X, \theta) | I_j, m] &= \pi_{j,m}^P(H_j, H_{-j}, X, \theta) - \varphi_{j,H_j}\end{aligned}$$

Model: Negotiation

- Key assumption: plan j 's expected profits from $H_j >$ expected profits from alternative network formed by reversing contract with h

$$E[\pi_{j,m}^{P,o}(H_j, H_{-j}, X, \theta) - \pi_{j,m}^{P,o}(H_j^h, H_{-j}, X, \theta) | Z_{j,m}] \geq 0$$

- form unconditional moments using positive-valued function of $Z_{j,m}$
- must be known to firms when they make their choice
 - use char in fixed cost and markup terms other than cost/admission
 - use indicators for some plan and market characteristics

Model: Negotiation

- ▶ Choose counterfactuals of reversing a single contract.
- ▶ Plans may respond by changing its response to other hospital's offers (passive beliefs rules out the following: plan responds to changes in h's offer by assuming other plans have different offers and therefore change their own networks)
- ▶ Two possible routes

Model: Negotiation

► Two possible routes

- Assume hospital can make an alternative offer to j that will prompt j to drop h from its network and not change its contract with other hospitals
- Allow plan to adjust its decisions wrt all other hospitals
 - find min of hospital h 's profits from all possible choices plan j can make in response to the deviation (given its other contracts and networks of other plans)
 - form inequality with difference between realized profit and minimized counterfactual profit

Results

- ▶ Estimate of θ for every specification is a singleton; could not satisfy all inequality constraints.
Why?
 - random disturbances
 - no. of moments used
 - no structural error (some component of profit function not observed by econometrician but used by the agent, that varies at p,h level)
- ▶ Comment
 - Inference for moment inequalities that find a set more complicated
 - Counterfactuals in the case of set identification?

Results: Substantive findings

- ▶ Hospitals in systems take a larger fraction of surplus, penalize plans that do not contract with all members
- ▶ Star hospitals capture high mark-ups
- ▶ Hospitals with higher costs/pt receive lower markups/pt

The Profit Inequality Approach

The Profit Inequality Approach

Pakes, Porter, Ho and Ishii (2010); Pakes (2010).

This approach allows for differences between the primitives that underly agents' decisions and the econometrician's constructs for those primitives.

We start with the measurement model that provides the relationship between these two objects.

The Profit Inequality Approach

Let our *observable* approximation to $\pi(\cdot)$

$$r^I(d, d_{-i}, z_i^o, \theta_0)$$

Define the error in this approximation as:

$$v(d, d_{-i}, z_i^o, z_i, \theta_0) = r^I(d, d_{-i}, z_i^o, \theta_0) - \pi(d, d_{-i}, z_i).$$

The Profit Inequality Approach

Then by definition:

$$r^I(d, d_{-i}, z_i^o, \theta_0) = \mathbb{E}[\pi(d_i, \mathbf{d}_{-i}, \mathbf{z}_i) | I_i] + v_{2,i,d}^I + v_{1,i,d}^I,$$

where

$$v_{2,i,d}^I = \mathbb{E}[v(d, d_{-i}, z_i^o, z_i, \theta_0) | I_i]$$

and

$$v_{1,i,d}^I = (\pi(d, \cdot) - \mathbb{E}[\pi(d, \cdot) | I_i]) + (v(d, \cdot) - \mathbb{E}[v(d, \cdot) | I_i]).$$

We will discuss the sources and interpretation of v_1^I and v_2^I in the following slides.

The Profit Inequality Approach

Note: For all $d \in D_i$, $\mathbb{E} [v_{1,i,d}^I | I_i] = 0$, by construction, while $\mathbb{E} [v_{2,i,d}^I | I_i] \neq 0$. It is this distinction that forces us to keep track of two separate disturbances and consider their relative importance.

The Profit Inequality Approach

Sources of v_1^I :

By construction v_1^I is the sum of:

1. Expectational error

$$\pi(d, ;) - E[\pi(d, .) | I_i]$$

and

2. Specification and measurement error

$$v(d, ;) - E[v(d, .) | I_i]$$

The Profit Inequality Approach

The expectational error is caused by:

- i uncertainty in $\mathbf{z}_i$, and/or
- ii asymmetric information which is uncertainty in $\mathbf{d}_{-i}$

So to compute its distribution (as we'd need to if we were using MLE, for example), we'd have to specify what each agent knew about its competitors, and then repeatedly solve for an equilibrium (a process which would typically require us to select among equilibria).

Since we'd have to compute returns from a counterfactual, the specification error would probably be non-trivial. The inequalities methodology does not require this step.

Sources of v_2^I and Selection:

v_2^I is that part of profits that the agent does condition on when making its decision but the econometrician has not included in the profit specification.

Since $v_{2,i}^I \in I_i$ and $d_i = d(I_i)$, d_i will generally be a function of $v_{2,i}^I$ (and perhaps also of $v_{2,-i}^I$).

This can generate a selection problem.

The Profit Inequality Approach

For example, temporarily ignore any difference between the agent's expectations (our $\mathbb{E}(\cdot)$) and the expectations generated by the true data generating process (our $E(\cdot)$).

Assume that x is an “instrument” in the sense that $\mathbb{E}(v_2^I|x) = 0$, and, in addition, that $x \in I$. Then:

$$\mathbb{E}(v_1^I|x) = \mathbb{E}(v_2^I|x) = 0.$$

The Profit Inequality Approach

However, these expectations do not condition on d_i

Any moment which depends on d_i (ie on the choice made by agent i - all our inequalities depend on this) requires properties of the disturbance conditional on d_i .

Since d is measurable given the information set I ,

$$\mathbb{E}(v_1^I | x, d) = 0.$$

The Profit Inequality Approach

However, since $v_2 \in I$, and

$$\mathbb{E}(\pi(\cdot)|\cdot) = \mathbb{E}(r(\cdot)|\cdot) - v_2,$$

if the agent chooses d^* then

$$v_{2,d} - v_{2,d^*} \geq \mathbb{E}(r(\cdot, d)|\cdot) - \mathbb{E}(r(\cdot, d^*)|\cdot)$$

so

$$\boxed{\mathbb{E}(v_{2,d}|x, d) \neq 0.}$$

More intuitively: firms are choosing d^* based on the observed v_{2,d^*} , so it makes sense that there's a selection issue, ie that $\mathbb{E}(v_{2,d^*}|x, d^*) \neq 0$.

The Profit Inequality Approach

This result implies that the statement that “ x is an instrument” does not “solve” the selection problem. Formally

$$E[\Delta\pi(d_i, \cdot)|x_i, d_i] = E[\Delta r(d_i, \cdot)|x_i, d_i] - E[v_{2,i,d_i}^I - v_{2,i,d'}^I|x_i, d_i] .$$

Theory gives us $E[\Delta\pi(d_i, \cdot)|x_i, d_i] \geq 0$, but this only implies

$$E[\Delta r(d_i, \cdot)|x_i, d_i] \geq 0$$

if

$$E[v_{2,i,d_i}^I - v_{2,i,d'}^I|x_i, d_i] \geq 0,$$

and *this requires additional conditions on the measurement model*. We will come back to these conditions shortly.

(See IC4.)

The Additional Requirements for Profit Inequalities

The Additional Requirements for Profit Inequalities

Recall that we need an assumption on the relationship between agents' expectations and the expectation operator generated by the data generating process, plus restrictions on the measurement model.

3. Agents' Expectations (IC3)

Let $h(\cdot)$ be a positive valued function. There is a known subset of the observed variables, say $x_i \in I_i$, that satisfy:

$$\frac{1}{N} \sum_i E[\Delta\pi(d_i, d', d_{-i}, z_i) | x_i] \geq 0 \Rightarrow$$
$$E \left[\frac{1}{N} \sum_i [\Delta\pi(d_i, d', d_{-i}, z_i) h(x_i)] \right] \geq 0.$$

The Additional Requirements for Profit Inequalities

Correct Expectations are Sufficient.

- ▶ Standard condition - each agent knows:
 - i the other agents' strategies ($\mathbf{d}_{-i}(I_{-i})$), and
 - ii the joint distribution of other agents' information sets and the primitive sources of uncertainty (of $(I_{-i}, \mathbf{z}_{-i})$) conditional on I_i , and regularity conditions.

- ▶ Weaker condition: agents' conditional expectations of the profit difference are correct.
 - This does not require knowledge of other agents' strategies, or the distribution of $(\mathbf{d}_{-i}, \mathbf{z}_{-i})$ conditional on I_i .
 - Agent uncertainty is permitted and we do not need to fully specify how the agent forms its expectations.

The Additional Requirements for Profit Inequalities

Incorrect Expectations are Possible.

All we need is the average of

$$\mathbb{E} [\Delta\pi(d_i, d', d_{-i}, z_i)|x_i] - E [\Delta\pi(d_i, d', d_{-i}, z_i)|x_i] \geq 0$$

Relevant cases:

- ▶ Agents' beliefs are not exactly right but the difference between agents' expectations on $\Delta\pi(., \theta_0)$ and the expectation of the data generating process are mean-zero conditional on x (in that case the expression is satisfied with equality). Or
- ▶ Agents can be “consistently overly optimistic” about the relative profits from the decisions they make.

The Additional Requirements for Profit Inequalities

4. **Condition on the Measurement Model (IC4)** This final condition is designed to deal with the selection problem noted above. Assume D_i is discrete. $\exists$ observed $x \in I_i$ and a function $c(\cdot) : D_i \times D_i \rightarrow R^+$ such that we satisfy:

$$E \left[\sum_{j \in D_i} \chi \{d_i = j\} c(j, d'(j)) \left(v_{2,i,j}^I - v_{2,i,d'(j)}^I \right) h(x_i) \right] \geq 0$$

where $\chi \{d_i = j\}$ is an indicator function.

Notes regarding IC4

- ▶ This expectation is an unconditional average (does not condition on d_i); for every possible $d \in D_i$ we specify a $d'(d)$.
- ▶ This average is an average in the *differences* in the $v_{2,i,j}^I - v_{2,i,d'(j)}^I$.
- ▶ Both (i) the weights and (ii) the comparison (d') can vary with j .

The Additional Requirements for Profit Inequalities

Satisfying IC4

► $\forall d, v_{2,i,d} = v_{2,i}.$

- This is Hansen and Singleton's (1982) classic article, but can allow for discreteness in choice sets, choices which are on the boundaries of the choice set, and interacting agents.

The Additional Requirements for Profit Inequalities

Satisfying IC4 (continued)

- ▶ $v_{2,i,d}$ can vary across decisions, but the same value of $v_{2,i,d}^I$ appears in more than one of them (so there are “group” effects).
 - Examples:
 - Entry models with location-specific fixed effects,
 - Social interaction models with group effects,
 - Panel data discrete choice models with choice-specific fixed effects, and
 - Cross-sectional discrete choice models where the same $v_{2,i,d}^I$ appear in more than one choice.
 - For each choice $d_i = j$, pick an alternative $d'(d)$ such that the two choices have the same value of $v_{2,i,d}$
 - e.g. pick an alternative in the same market; the same group; a firm making the same choice in a different time period.
 - The v_2 terms will difference out when we generate the inequality.

Satisfying IC4 (continued)

- ▶ Ordered choice models (including the vertically-differentiated demand model used in I.O.).
 - E.g. the firm is buying a discrete number of units, so $d_i \in Z_+$ and $v_{2,i}$ is a cost component known to the agent but not the econometrician.
 - Take $d'(j) = j + 1$. The difference in profits will always contain v_2 , now interpreted as the cost savings from not purchasing the additional unit.
 - We will take an unconditional average across this cost (not conditional on the choice).
 - This is the method used in Ishii (2005).

Satisfying IC4 (continued)

- ▶ Contracting models in which v_2 is interpreted as a component of the contract that the agents know but the econometrician does not.
 - The cost is a profit to the seller if the contract is established and a saving to the buyer if the contract is not accepted.
 - We can difference it out by adding together the inequality of the buyer and that of the seller.
 - This is an extension of the model in Ho (2009); the original paper takes account of v_1 but ignores v_2 .

Satisfying IC4 (continued)

- ▶ Models for micro data where a variable needed for an inequality is unobserved (or is measured with error) at the micro level but is observed at a higher level of aggregation (say because of the availability of Census data).

Estimation from Profit Inequalities

Estimation from Profit Inequalities

C1 and C2 imply that for each i

$$0 \leq \sum_j \left[(\chi \{d_i = j\} c(j, d'(j)) \Delta \pi(j, d'(j), .)) h(x_i) \right].$$

Average over i . IC3 implies that this average is less than or equal to

$$E \left[\frac{1}{N} \sum_i \sum_j (\chi \{d_i = j\} c(j, d'(j)) \Delta \pi(j, d'(j), .)) h(x_i) \right],$$

which from IC4 is less than

$$E \left[\frac{1}{N} \sum_i \sum_j (\chi \{d_i = j\} c(j, d'(j)) \Delta r(j, d'(j), .)) h(x_i) \right].$$

Since this last inequality is in terms of *observable* moments, we can use its sample analogue as a basis for estimation.

Estimation from Profit Inequalities - Katz (2007)

Katz's problem was to estimate the costs shoppers assign to driving to a supermarket.

These costs are important for the choice of supermarket locations and, as a result, for the analysis of the impact of zoning regulations.

They have been difficult to analyze empirically with standard choice models because of the complexity of the choice set facing consumers (all possible bundles of goods at all supermarkets).

Estimation from Profit Inequalities - Katz (2007)

Assume that the agents' utility functions are additively separable functions of:

- ▶ the utility from the basket of goods the agent buys,
- ▶ expenditure on that basket, and
- ▶ drive time to the supermarket.

Since utilities are only defined up to a monotone transformation, there is a free normalization for each individual. Katz normalizes the coefficient on expenditure for each individual to equal one.

He allows for heterogeneity in the cost of drive time that is known to the agents when they make their decision but unobserved by the econometrician. This will be one component of $v_{2,i}$.

Estimation from Profit Inequalities - Katz (2007)

Possible counterfactuals: Purchase of any bundle of goods at any store.

For a particular d_i choose $d'(d_i)$ to be the purchase of

- ▶ the same basket of goods
- ▶ at a store which is further away from the consumer's home than the store the consumer shopped at.

Estimation from Profit Inequalities - Katz (2007)

This choice of $d'(d_i)$ allows us to difference out the impact of the basket of goods chosen on utility.

I.e. if $e(d)$ and $dt(d)$ provide the expenditure and the drive time for store choice d , and $(\theta + v_{2,i})$ is agent i 's cost of drive time (in units of expenditure),

$$\begin{aligned} E \left[\sum_j \chi \{d_i = j\} \Delta \pi(j, d'(j), z_i) | I_i \right] = \\ E \left[\sum_j \chi \{d_i = j\} (e(j) - e(d'(j)) + (\theta + v_{2,i}) (dt(j) - dt(d'(j)))) | I_i \right] \geq 0, \\ \text{at } \theta = \theta_0. \end{aligned}$$

Assuming, as seems reasonable, that $(dt(d_i), dt(d'(d_i))) \subset I_i$, this together with the fact that $dt(j) - dt(d'(j)) < 0$ by choice of alternative implies that:

$$\mathbb{E} \left[\sum_j \chi \{d_i = j\} \left(\frac{e(j) - e(d'(j))}{dt(d'(j)) - dt(j)} - (\theta + v_{2,i}) \right) | I_i \right] \leq 0.$$

Let θ be the average of the cost of drive time across consumers, so $\sum_i v_{2,i} = 0$ by construction, and assume IC3. Then:

$$E \left[\frac{1}{N} \sum_i \left(\frac{e(d_i) - e(d'(d_i))}{dt(d'(d_i)) - dt(d_i)} \right) \right] - \theta \leq 0, \text{ at } \theta = \theta_0.$$

This result provides a lower bound to θ .

Were we to consider a second alternative in which the bundle of goods purchased was the same as in the actual choice but the counterfactual store required *less drive time*, we would also get an upper bound to θ_0 .

Katz (2007) shows that these bounds are quite informative and provides a range for the average cost of drive time which accords with auxiliary information, while more standard discrete choice estimators do not.

Estimation from Profit Inequalities - Katz (2007)

To obtain these inequalities we chose an alternative that allowed us to difference out the impact of the bundle of goods chosen on utility (differencing out our "group" effect).

We then rearranged these differences to form a moment which was linear in the remaining source of v_2 variance no matter d_i (the source being differences in the costs of travel time).

Estimation from Profit Inequalities - Katz (2007)

If we were interested in the impact of a particular good purchased on utility, we would have considered baskets of goods which differed only in that good and goods which had cross partials with that good in the utility function, at the *same* supermarket (thus differencing out the effects of travel time and other components of utility).

A lot more options would present themselves were we to have data on multiple shopping trips for each household.

Summary

Full Information - No Error Model

- ▶ Does not allow for:
 - Specification or measurement error
 - Asymmetric or incomplete information
 - Incorrect expectations

except to the extent that these details do not cause differences in the profits earned from different choices.

- ▶ Requires a parametric assumption on the distribution of v_2 .

Profit Inequalities Model

- ▶ Allows for specification errors, incorrect expectations, and incomplete and asymmetric information.
- ▶ The econometrician does not need to specify what the agent knows about either its competitors, or about the state of nature.
- ▶ However, it requires a (sometimes tricky) restriction on v_2 .
 - Given that restriction, there is no need for a distributional assumption here.